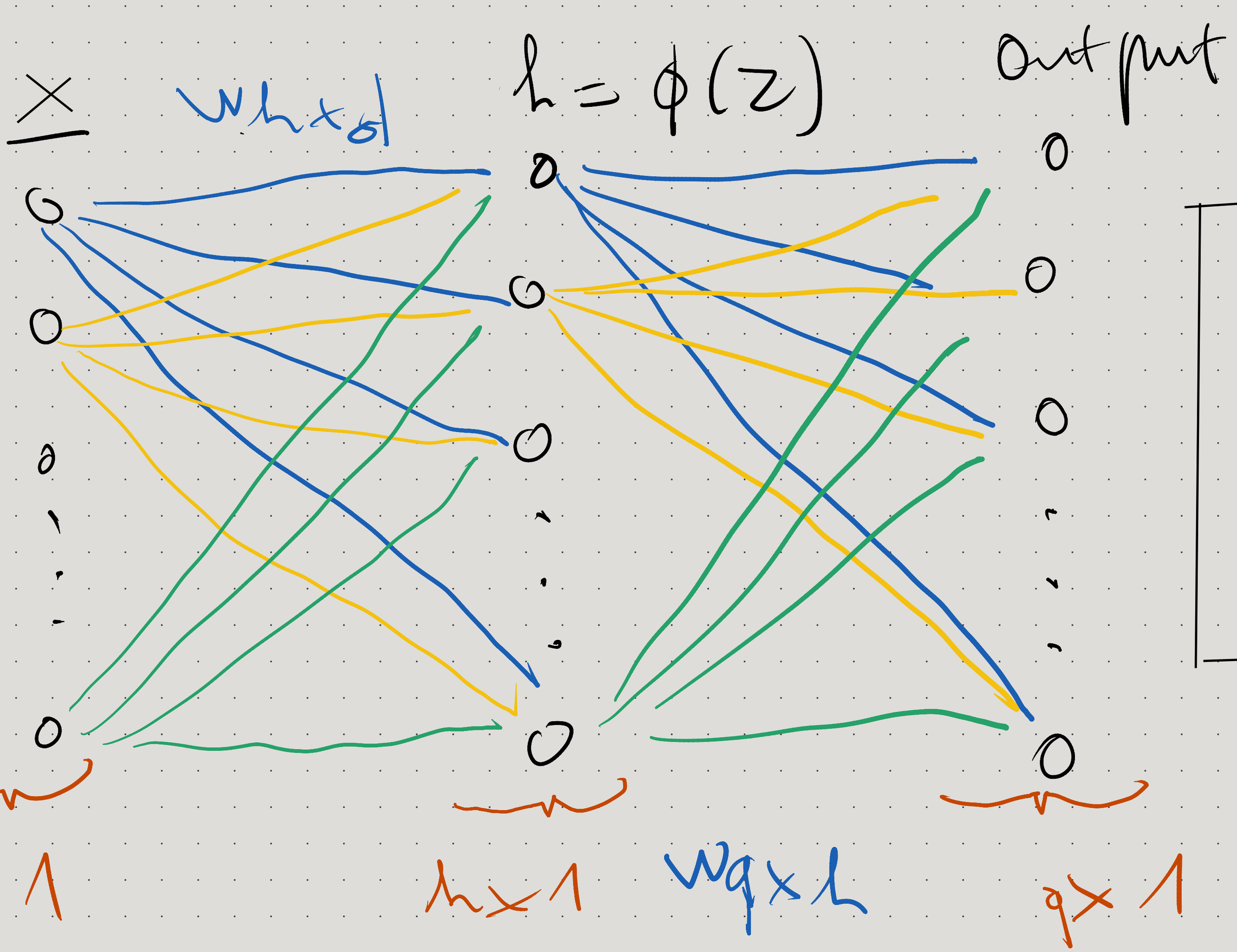
Forward Propagation, backward propagation & computational graphs:

orward propagation:


X
what
h = ϕ(z)
output
0
0
0
0
0
0
0
0
0
0
0
0
0
1
1
1

* Loss :

* Formula :

$$z = W^{(1)} x$$

$$h = \phi(z)$$

$$o = W^{(2)} h$$

$$L = \ell(o, y)$$

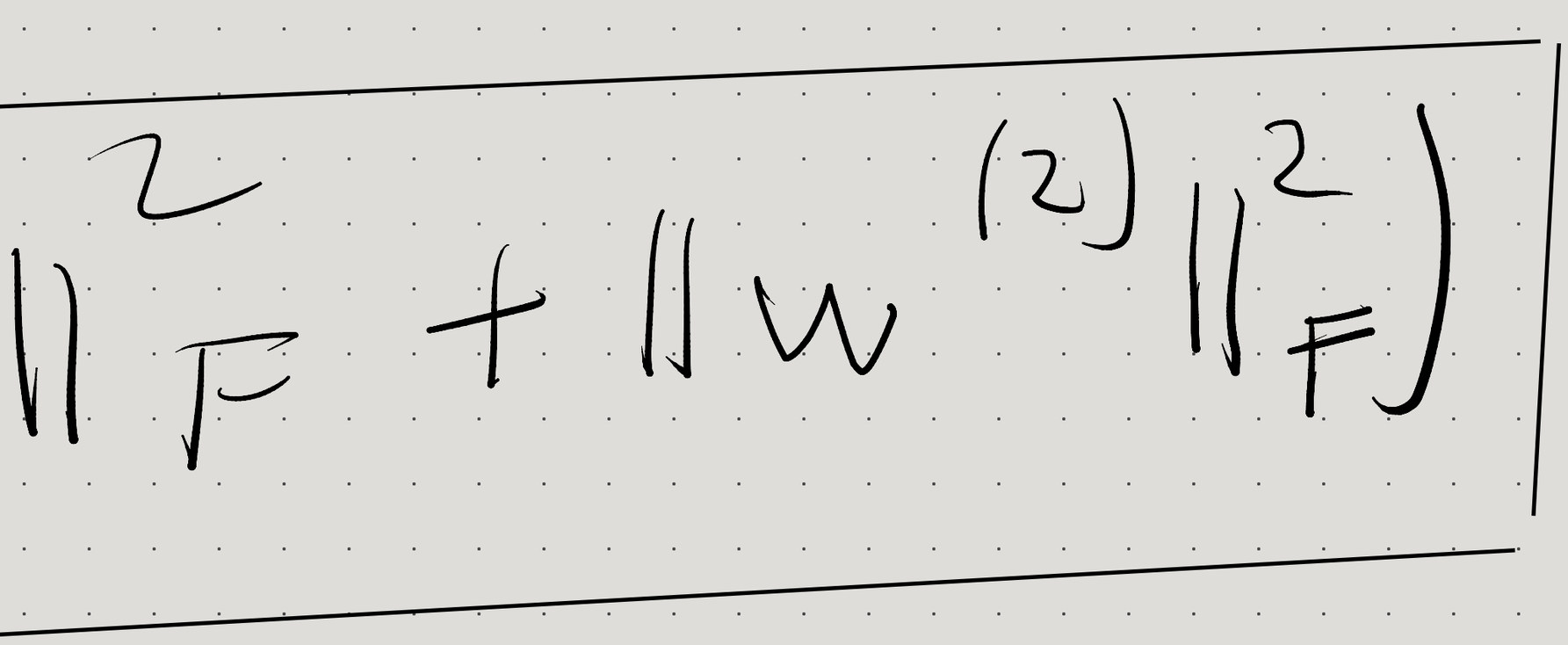
* Weight decay

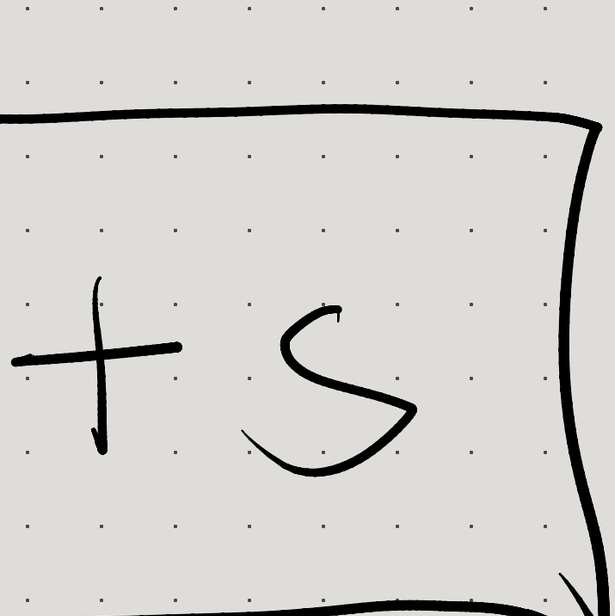
$$s = \frac{\lambda}{2} \left(\|W^{(1)}\| \right.$$

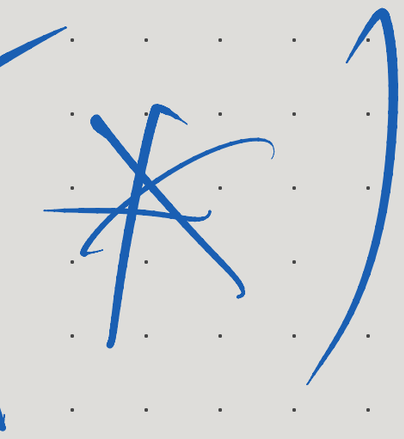
$$\Rightarrow \text{Final loss : } J = L$$



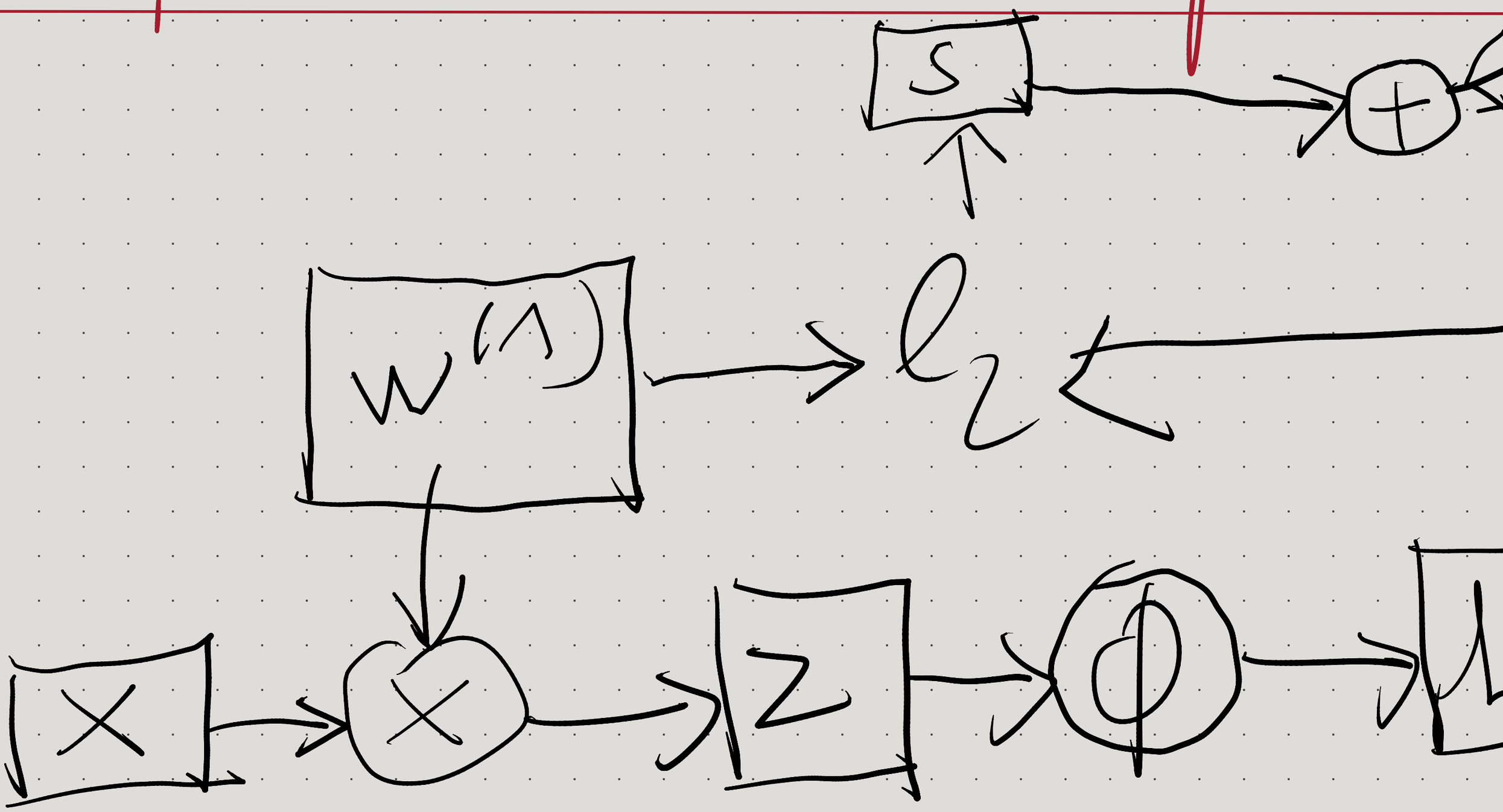








Computational Graph of

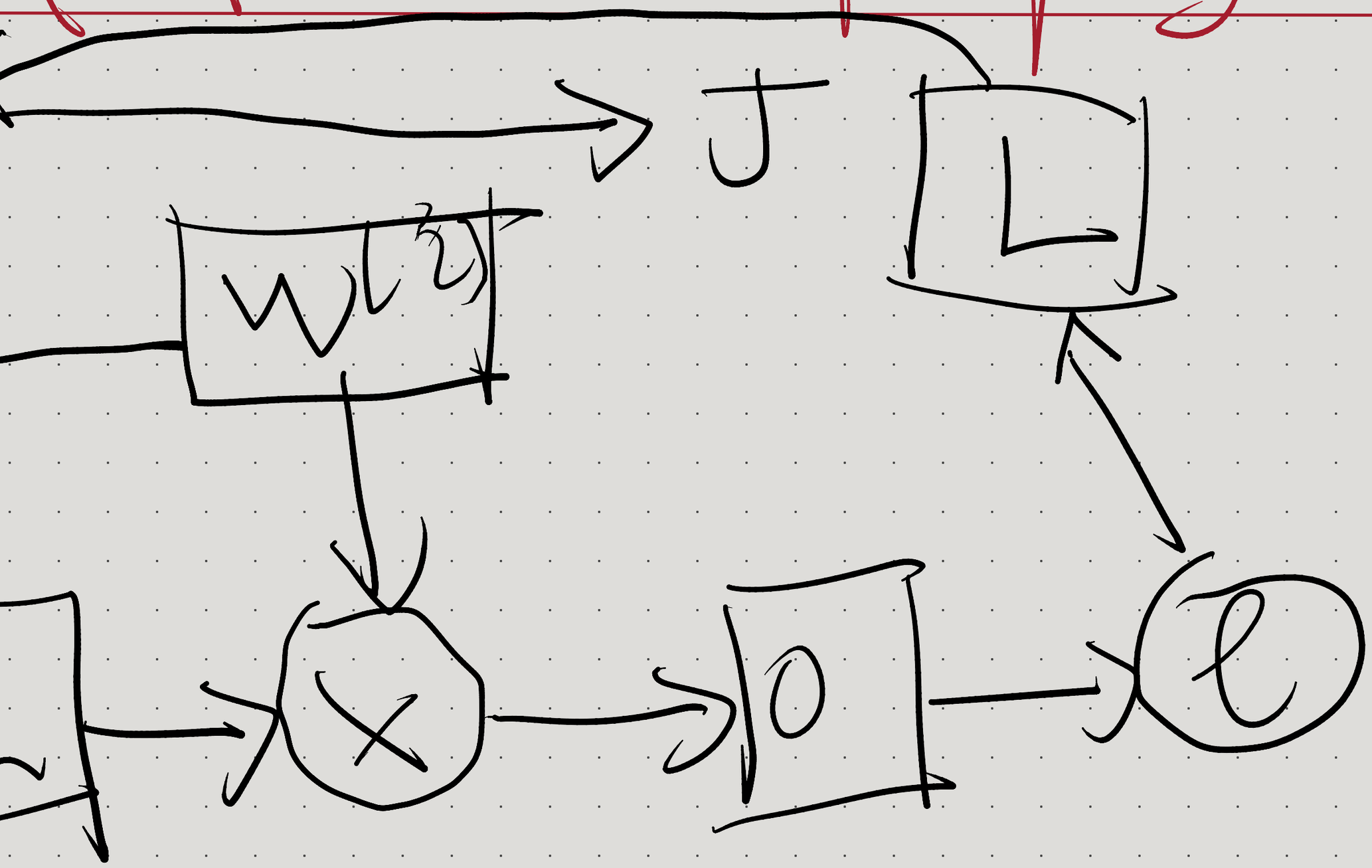


Backpropagation:

* Gradient of neural netw

$$\frac{dJ}{dW^{(2)}} = \frac{\partial J}{\partial L} \cdot \frac{\partial L}{\partial O}$$

forward propagation:

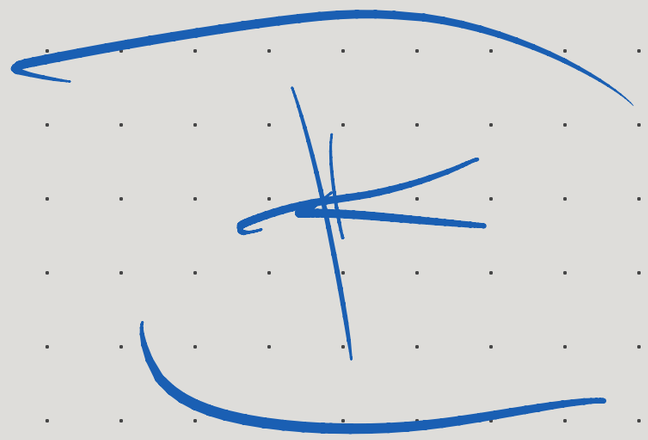


ork params.

$$\frac{\partial J}{\partial W^{(2)}} = \frac{\partial J}{\partial S} \cdot \frac{\partial S}{\partial}$$

$$\frac{\partial J}{\partial W^{(2)}} =$$





$$= 1 \cdot \frac{\partial L}{\partial o} \cdot \frac{\partial o}{\partial h^{(1)}}$$

$$\frac{\partial J}{\partial w^{(1)}} = \frac{\partial J}{\partial L} \cdot \frac{\partial L}{\partial o} \cdot \frac{\partial o}{\partial h^{(1)}} \cdot \frac{\partial h^{(1)}}{\partial w^{(1)}}$$

$$= \frac{\partial L}{\partial o} \cdot w^{(2)} \cdot \frac{\partial h^{(1)}}{\partial w^{(1)}}$$

$$+ 1 \cdot \lambda w$$

$$+ \lambda w \cdot 0$$

$$\frac{\partial o}{\partial h} \cdot \frac{\partial h}{\partial z} \cdot \frac{\partial z}{\partial w}$$

$$\phi(2) \cdot x^{T} +$$

$$\frac{\partial S}{\partial w^{(1)}} + \frac{\partial S}{\partial w^{(1)}}$$

$$\frac{\partial S}{\partial w^{(1)}} \quad (*)$$

(*) Vani



(*) Gradient W

$\frac{\partial o}{\partial}$ →

Shrinking & Explodin

$$= f_\ell\left(h^{(\ell-1)}\right)\Big] \Longrightarrow$$

(ℓ) as follows:

$$\frac{\partial h^{(L)}}{\partial h^{(L-1)}} \cdot$$

2 gradients:

$$\nabla = f_L \circ f_{L-1} \circ \cdots \circ$$

$$\cdots \circ \frac{\partial h^{(l+1)}}{\partial h^{(l)}} \cdot \frac{\partial h^{(l)}}{\partial h^{(l-1)}} \cdot \cdots \cdot \frac{\partial h^{(1)}}{\partial W^{(1)}}$$



$f_1(x)$

$$\frac{\partial L}{\partial w^{(e)}} \rightarrow$$

$$\times \frac{\partial L}{\partial w^{(e)}} = \frac{\partial L}{\partial (w^{(e)})^{(l)}} \frac{\partial (w^{(e)})^{(l)}}{\partial w^{(e)}} \rightarrow$$

$$\frac{\partial L}{\partial (w^{(e)})^{(l)}} \rightarrow$$

$$\frac{\partial L}{\partial (w^{(e)})^{(l-1)}} \rightarrow$$

$$\frac{\partial L}{\partial (w^{(e)})^{(0)}}$$

→ Vanishing

(*) Vanishing gradient

$\delta h^{(L-1)}$ $\delta h^{(L-1)}$

$M^{(L)}$ $M^{(L-1)}$

$\delta_0^{(L)} = \delta L \times M^{(L-1)}$

Exploding Gradients

δh δW

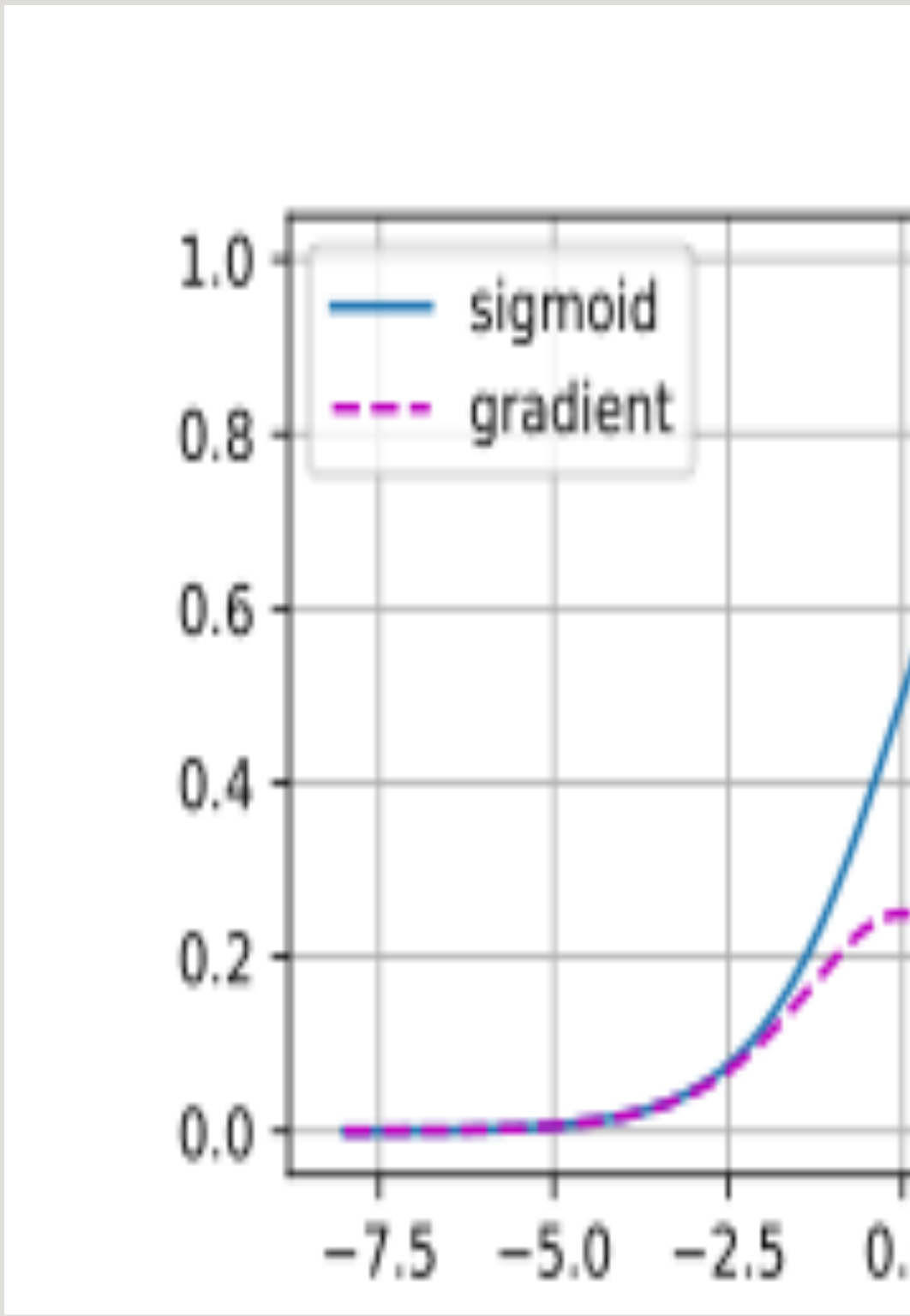
$M^{(l+1)}$ $V^{(l)}$

$L)$ $M^{(L-1)}$ $\cdots$ $M^{(l+1)}$

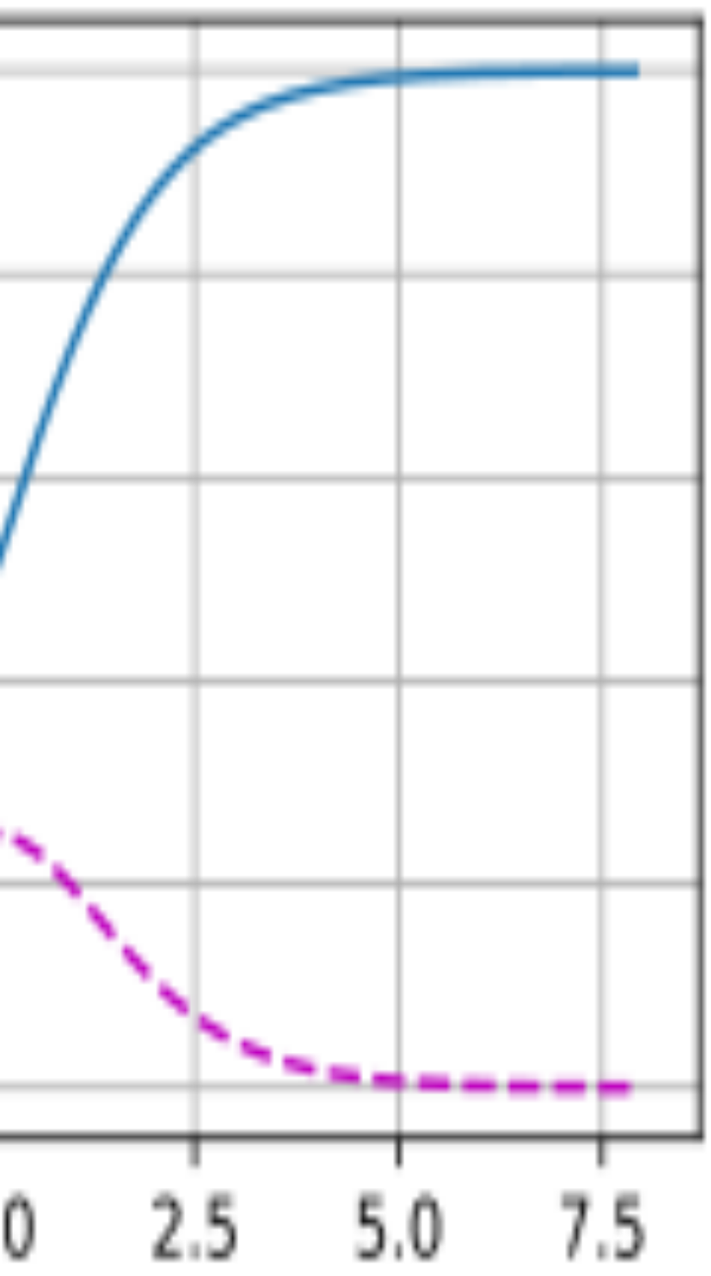
$\cdots$

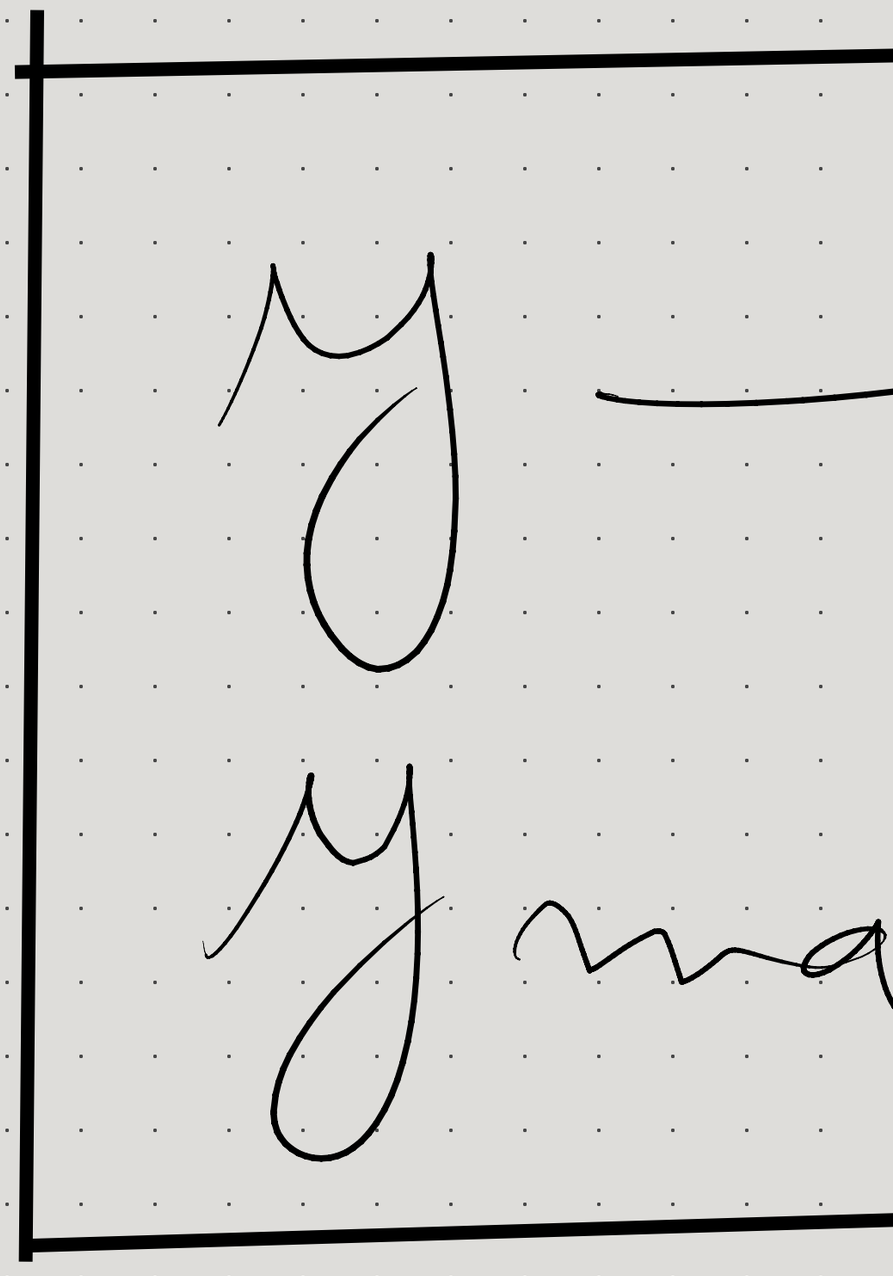
$\frac{\partial C}{\partial}$

-1) . V(e)


1.0
0.8
0.6
0.4
0.2
0.0
sigmoid
gradient
−7.5
−5.0
−2.5

Sigma





Gradient

oi d fonction

→ 0

$t = 0{,}25$

t vers $+\infty$

Relu fu

- ReLU(x) =

Gradient Re

(*) Exploding G

Happen when

nction :

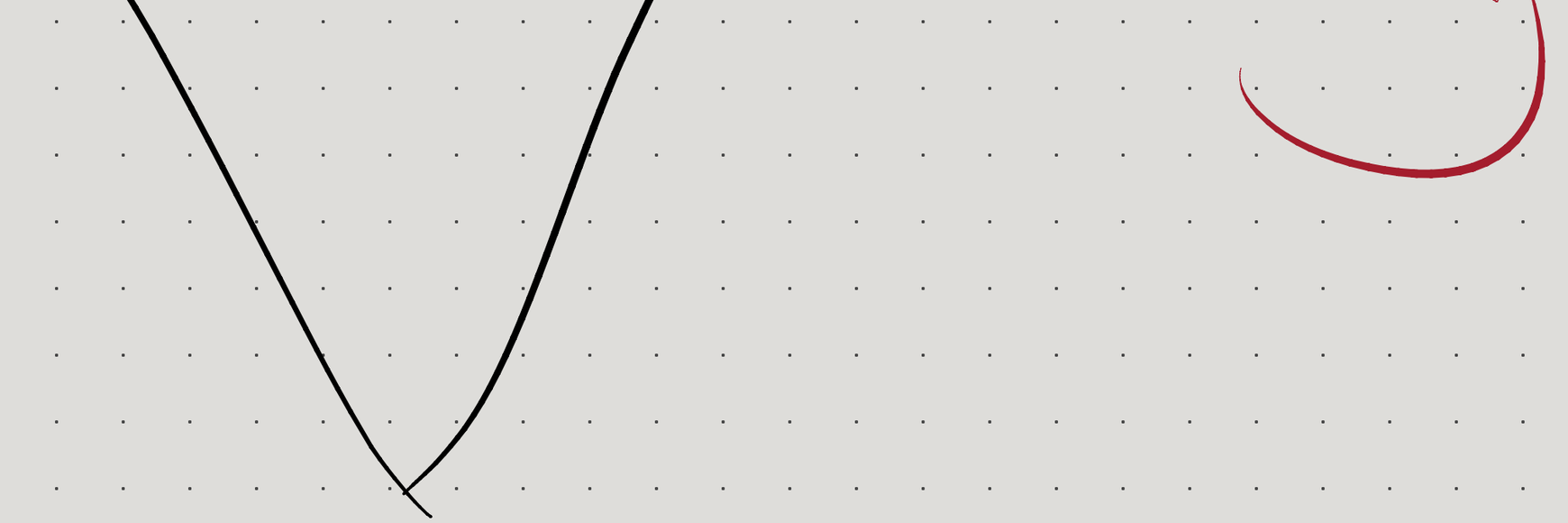
$$\max(0, x)$$

$$\ell U = 1$$

Gradient :

multivariable math





(*) Breaking the

- Randomize for b

- Drop out regu

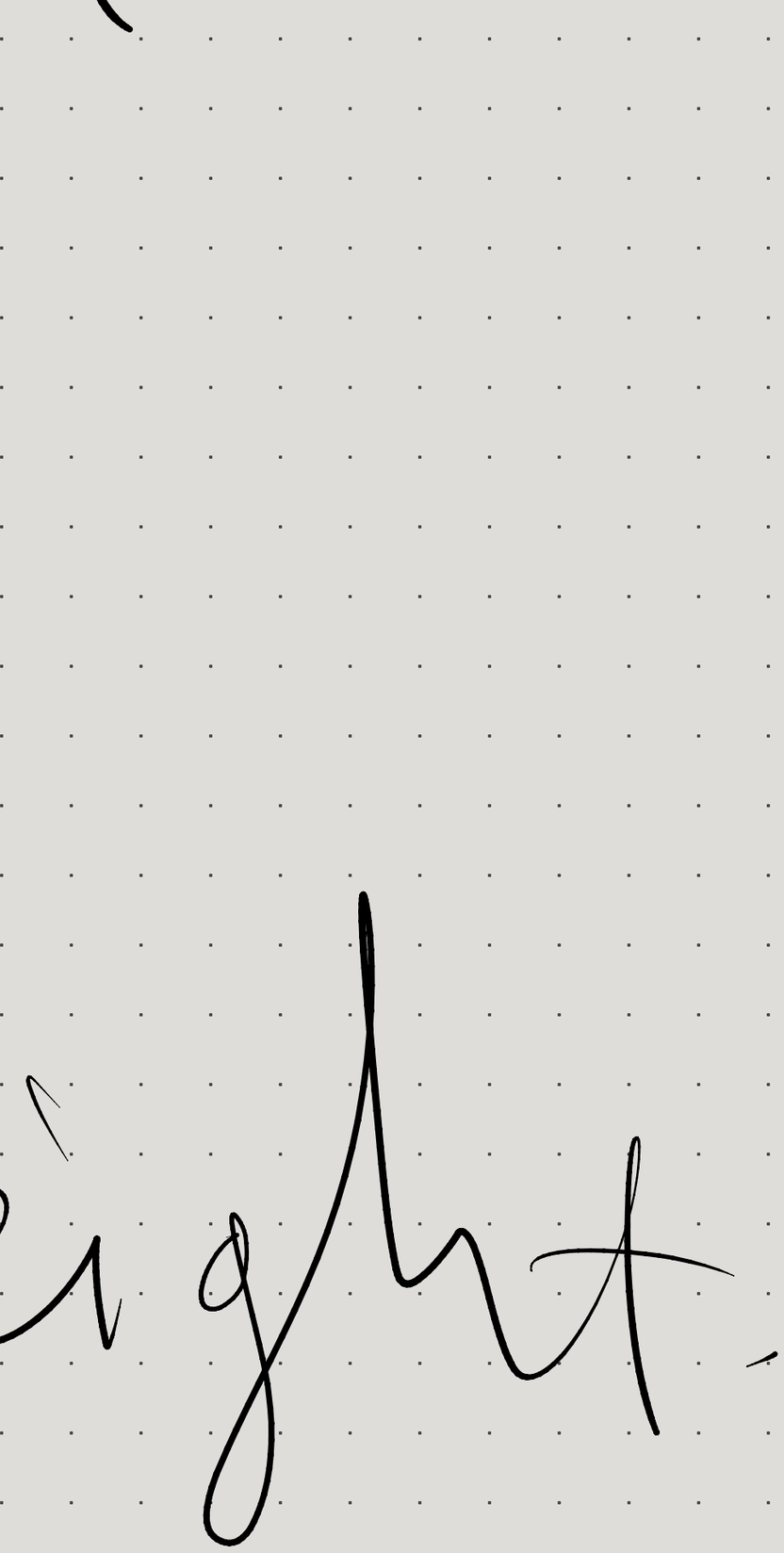
(*) Parameter I

Xavior initializat

Symmetry breaking Same ve

larization.

initializations



+) Normal distribution

$$W \sim N(0, \frac{2}{n_{in}})$$

+) Uniform distribution

$$W \sim U(-\sqrt{\frac{6}{n_{in}+n_{out}}}, \sqrt{\frac{6}{n_{in}+n_{out}}})$$

tion

tmont

bution

6

yintmat

6

Jmintn

